Hydrogen energy four ways

This is the traditional formula for E_n.

$$E_n = -\frac{\mu}{2n^2}\left(\frac{e^2}{4\pi\varepsilon_0\hbar}\right)^2 \tag{1}$$

The following formula eliminates elementary charge e.

$$E_n = -\frac{\hbar^2}{2n^2\mu a_0^2} \tag{2}$$

The following E_n can be divided by $\hbar$ to obtain angular frequency ω.

$$E_n = -\frac{\alpha\hbar c}{2n^2 a_0} \tag{3}$$

The following E_n is irreducible.

$$E_n = -\frac{\alpha^2\mu c^2}{2n^2} \tag{4}$$

Equation (1) reduces to (4) by substituting

$$e^2 = 4\pi\varepsilon_0\alpha\hbar c$$

Equations (2) and (3) reduce to (4) by substituting

$$a_0 = \frac{\hbar}{\alpha\mu c}$$

Eigenmath script